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EDUCATION AND RESEARCH EXPERIENCE

PhD – Computational Chemistry

[October 2022 – September 2026]

Cardiff University - Cardiff, United Kingdom

Thesis: Controlling Selectivity of Fischer-Tropsch Catalysts Using Mn Promoters - A Computational Modelling Study

- Investigated the stability of CoO and Co₃O₄ surfaces and nanoparticles using **ab initio thermodynamics**, providing insights into the structural stability of Fischer-Tropsch catalysts.
- Developed a **machine learning-assisted DFT workflow** to model CoO_x/TiO₂ interfaces, enabling efficient investigation of metal-support interactions in catalyst systems.
- Elucidated the role of Mn and Ru promoters in Fischer-Tropsch catalysts using **density functional theory (DFT)** to study Co-Mn-Ru interactions on TiO₂.
- Identified the nature of active surface sites using **first-principles CO vibrational frequency analysis** and enabled improved understanding of Fischer-Tropsch catalyst surfaces.

Research Impact and Collaboration

- Collaborated with more than 10 experimental researchers across academia and industry on the research project to support catalyst design and interpretation of experimental data.
- Presented research findings to university collaborators and industrial scientists through quarterly project meetings.
- Developed expertise in communicating complex computational results to both computational and experimental audiences.

Industrial Placement

[April 2026 - May 2026]

bp industries - Sunbury, United Kingdom

Business Report: Machine Learning discovery of CoMn catalyst descriptors

- Trained **machine learning (ML) models** (Random Forest, Gradient Boosting, and Neural Networks) to identify the key factors controlling Fischer-Tropsch catalyst activity and selectivity using large experimental datasets.
- Applied **explainable AI techniques** (SHAP and ICE) to interpret model predictions and provide chemically meaningful insights for catalyst design.
- Implemented a **Bayesian optimization workflow** to identify new operating conditions with improved selectivity towards desired Fischer-Tropsch products.

Research Impact and Collaboration

- Produced a business report summarising technical findings, recommendations, and future research opportunities for industrial stakeholders.
- Developed clean, modular, and well-documented code that was transferred to industrial researchers for future use and further development.

- Built relationships with teams across Licensing & Patents, Quantum Computing, Digital Science, and AI Ethics to understand how scientific research is translated into commercial value and industrial strategy.

Master of Science – Chemistry

[September 2020 - July 2022]

Indian Institute of Technology (IIT) Madras - Chennai, India

Grade: 9.18 CGPA (out of 10.00)

Thesis: Anion- π Interaction - Theoretical study of its possible role in functionalized fullerene-based catalysis

- Explored the role of the counter-intuitive anion- π interaction in stabilizing reaction intermediates and how these interactions are manifested in the transition states.
- Probed the nature and strength of the anion- π interaction using various computational methods such as Natural Bonding Orbital (NBO) analysis and energy decomposition analysis (EDA)
- Elucidated the energetics of the reaction and provided a rationale for the design of a novel catalyst based on our analysis.

Bachelor of Science (Honors) – Chemistry

[July 2017 - June 2020]

The Maharaja Sayajirao University of Baroda - Vadodara, India

Grade: 9.01 CGPA (out of 10.00)

All India Senior School Certificate Examination

Gujarat Public School - Vadodara, India

Percentage: 89.2%

QUANTUM COMPUTING EXPERIENCE

National Quantum Computing Centre (NQCC) Quantum Computing Hackathon 2026

Problem: Forecasting Seizures from Brain Electrical Activity | Industry Partner: Applied Quantum Computing (AQC)

- Trained **Variational Quantum Classifier (VQC)** and **Quantum Neural Network (QNN)** on Quantinuum's H2 emulator and tested the model performance on IBM's Heron r2 quantum processor.
- Evaluated the **performance and scaling behaviour** of QNN classifier relative to classical ML algorithms.
- **Implemented the workflow** to obtain benchmark performance metrics for classical ML algorithms.

National Quantum Computing Centre (NQCC) Quantum Computing Hackathon 2025

Problem: Maximum Independent Set (MIS) | Industry Partner: BT

- Developed a graph-to-hardware mapping algorithm to embed graph nodes onto a 2D neutral-atom array while satisfying the Rydberg blockade constraints for connected vertices.
- Performed **Analog Hamiltonian Simulations (AHS)** on **QuEra** neutral-atom quantum hardware via Amazon Braket.
- Contributed to the **implementation and optimisation** of quantum algorithms for solving the Maximum Independent Set problem on QuEra's quantum hardware.

National Quantum Computing Centre (NQCC) Quantum Computing Hackathon 2024 (*Silver Medal*)

Problem: Minimum Vertex Cover (MVC) | Industry Partner: BT

- Implemented the **QUBO Hamiltonian** formulation for the Minimum Vertex Cover problem.
- Implemented a **Quantum Approximate Optimisation Algorithm (QAOA)** implementation using the **Amazon Braket** local simulator, based on published literature and reference implementations.
- Evaluated the performance and scaling behaviour of QAOA relative to **D-Wave's quantum annealer** and hybrid solver.
- Gained practical experience with quantum annealing, gate-based quantum computing, and quantum optimisation workflows.

PUBLICATIONS

1. Panneerselvam Murugesan, **Akash Hiregange**, and Archita Patnaik. "Reciprocity of $C=O \cdots \pi$ interactions with the dominant anion- π on fullerene (C60)-amine-based organocatalysts: a mechanistic elucidation for addition vs. decarboxylation reaction." *Physical Chemistry Chemical Physics* 25, no. 15 (**2023**): 10647-10660.
2. Chaudhari Amit, Stishenko Pavel, **Akash Hiregange**, Christopher R. Hawkins, Misbah Sarwar, Stephen Poulston, and Andrew J. Logsdail. "Ab initio insights into support-induced sulfur resistance of Ni-based reforming catalysts." *Catalysis Science & Technology* 16, no. 4 (**2026**): 1457-1472.
3. **Akash Hiregange**, Pavel Stishenko, Sarah J. Haigh, James Paterson, Mark Peacock, Andrew J. Logsdail. "First-Principles investigation of surface reconstructions of polar CoO (111) surface and its effect of nanoparticle morphology" Submitted to *Chemistry of Materials* (**2026**)
4. Qizhen Li, **Akash Hiregange**, Matthew Lindley*, Fred Tinkamanyire, Mateusz Salamon, Pavel Stishenko, Joseph Parker, Mark Peacock, Christopher Hardacre, James Paterson, Alex S. Walton, Andrew J. Logsdail*, Sarah J. Haigh*. "Metal-Support Interactions of Mn-promoted Co/TiO₂ Catalysts" Submitted to *Advanced Materials* (**2026**)

GRANTS AND FUNDING

Secured approximately £44,000 in competitive research funding and computational resources as principal proposer during my PhD, including national supercomputing allocations, research collaboration funding, and undergraduate research internship awards.

ARCHER2 UK National Supercomputing Service [2022–2026]

Successfully secured competitive allocations of ARCHER2 computational resources on eight occasions during my PhD, totalling approximately 180,000 Compute Units (CU) (equivalent value of approximately **£35,100** at £0.20/CU). All proposals were developed from my PhD research and enabled large-scale atomistic simulations.

bp-ICAM Researcher Collaboration Fund, bp-ICAM [2026]

Data-Driven Exploration of Co-Mn-Ru Cluster Formation on TiO₂ Surfaces for Fischer-Tropsch Catalysis - **£4,148.**

Conceived and led the proposal to establish collaboration between two ICAM research projects, facilitating knowledge exchange and accelerating data-driven materials discovery.

On-Campus Internship Scheme, Cardiff University [2026]

Machine learned (ML) approach to understand the stability of cobalt nanoparticles over different particle size ranges - **£2,017.**

Conceived and led the proposal for a project to support a funded summer research placement for an undergraduate student.

On-Campus Internship Scheme, Cardiff University [2025]

Modelling CO adsorption on cobalt metal using machine-learned (ML) force fields - **£2,760**.

Conceived and led the proposal for a project to support a funded summer research placement for an undergraduate student.

TEACHING AND RESEARCH SUPERVISION EXPERIENCE

Teaching

Undergraduate Laboratory Demonstrator, Cardiff University [2022–2026]

- Demonstrated undergraduate chemistry and computational laboratory classes, providing guidance on experimental techniques, data analysis, and Python programming for computational practicals.
- Supported students in developing problem-solving skills and understanding of theoretical concepts during laboratory sessions.
- Contributed to the assessment and marking of laboratory reports and practical exercises.

Research Supervision

MChem Research Project Supervisor [2025–2026]

- Conceived an MChem research project entitled *First-principles investigation of hydrogen diffusion on Ru-doped Co surfaces for Fischer-Tropsch catalysis*.
- Supervised the student throughout the project, providing training in computational chemistry methods, data analysis, scientific writing, and thesis preparation.
- Contributed to the assessment of the completed project.

Summer Research Placement Supervisor [2022–2026]

- Supervised four undergraduate summer placement students (one annually), each undertaking an 8-week research project.
- Designed research projects, provided day-to-day scientific guidance, and trained students in atomistic simulations, machine learning methodologies, and scientific communication.

SCHOLARSHIPS, AWARDS, AND PRIZES

Principal Avasare Gold Medal (2020) - Awarded for securing 1st rank in the BSc Chemistry graduating class.

Merit Scholarship (2019, 2021, 2022) - Awarded for outstanding academic performance during the final year of the BSc programme and both years of the MSc programme.

Poster Prize, Cardiff Catalysis Institute Conference (2025) - Awarded for an outstanding research poster presentation.

Poster Prize, MS University Science Open House (2020) - Awarded for an outstanding research poster presentation.

Silver Medal, National Quantum Computing Centre (NQCC) Quantum Computing Hackathon, University of Warwick (2024) - Member of the award-winning team recognised for developing an innovative quantum computing solution during the national hackathon.

TECHNICAL SKILLS AND COMPETENCIES

TECHNICAL SKILL	SOFTWARE/PACKAGES
Electronic Structure Calculations (DFT)	FHI-aims, GPAW, Gaussian09
Machine-learning Interatomic Potentials	MACE, GAP
Molecular Dynamics and Monte-Carlo Simulations	Atomic Simulation Environment (ASE), SuSMoST
Programming	Python, Bash, R
Quantum Computing	Software: Qiskit, Amazon Braket, Nexus (Quantinuum), Ocean SDK (D-Wave), PennyLane Hardware: D-wave, QuEra, Quantinuum, IBM
High-Throughput Computational Workflows and Automation	ASE, Pymatgen, Bash, Slurm
Machine-Learning and AI	PyTorch, TensorFlow, Keras, Scikit-Learn
Model Interpretation & Explainable AI	SHAP, Individual Conditional Expectation (ICE)
Version Control	Git, Github
Operating systems	Linux, Windows
High-Performance Computing	Slurm, OpenPBS, Linux-based HPC environments
Data Analysis and Visualization	Python, OVITO, POV-Ray, VESTA, VMD, Origin
Scientific Visualisation & Graphics	Blender
Scientific Documentation	LaTeX, Microsoft Office

ORAL PRESENTATIONS

- **2026 – “Deciphering Metal-Support interactions in Fischer-Tropsch catalysts: Atomistic modelling of CoO_x/TiO₂ interfaces”** - UK Catalysis Conference, Loughborough University, UK.
- **2026 – “Active Learning-Driven Exploration of Cobalt Phase Stability in FT Catalysts”** - Computational Chemistry Seminars, Cardiff University, UK.
- **2025 – “Controlling selectivity of catalysts through the interaction of cobalt with promoters”** - bp-ICAM Annual Conference, Manchester, UK.
- **2025 - "Computational insights into the stability and phase transition of cobalt oxide nanoparticles for Fischer-Tropsch catalysis"** - Computation for Applied Catalysis (CAC) Meeting, University of Leeds, UK.
- **2024 - "Computational insights into the stability and phase transition of cobalt oxide nanoparticles for Fischer-Tropsch catalysis"** - FHI-aims Users and Developers Meeting, University of Warwick, UK.
- **2024 - "Understanding the nature of Fischer-Tropsch catalyst surface using first-principles CO vibrational frequency analysis"** - Materials Chemistry Consortium Annual Conference, Daresbury Laboratory, UK.
- **2024 - "Computational insights into the stability and phase transition of cobalt oxide nanoparticles for Fischer-Tropsch catalysis"** - Computational Molecular Science (CMS) Conference, University of Warwick, UK.

- **2024 - “Controlling selectivity of catalysts through the interaction of cobalt with promoters”** - bp-ICAM Annual Conference, Manchester, UK.
- **2023 - “Capturing the ground state electronic and magnetic properties of Cobalt containing metal and mixed metal oxides using density functional theory (DFT)”** - Computational Chemistry Seminars, Cardiff University, UK.

POSTER PRESENTATIONS

- **2026 – “Deciphering Metal-Support interactions in Fischer-Tropsch catalysts: Atomistic modelling of CoO_x/TiO₂ interfaces”** - Materials Chemistry Consortium Annual Conference, Daresbury Laboratory, UK.
- **2025 – “Deciphering Co-Ru-Mn interactions in Mn-Ru-promoted Fischer-Tropsch catalysts”** - bp-ICAM Annual Conference, Manchester, UK.
- **2025 - "Insights into the stability and phase transition of cobalt oxide nanoparticles for Fischer-Tropsch catalysis"** - 17th International Conference on Materials Chemistry, Edinburgh, UK.
- **2025 - "Modelling CoO_x/TiO₂ interfaces: A MACE accelerated DFT study"** - RSC Theoretical Chemistry Graduate Meeting 2025 – Virtual.
- **2025 - "Stability and Phase Transition of Cobalt Oxide Nanoparticles for Fischer-Tropsch Catalysis"** - Cardiff Catalysis Institute (CCI) annual conference, Cardiff University, UK.
- **2024 - "Computational insights into the stability and phase transition of cobalt oxide nanoparticles for Fischer-Tropsch catalysis"** - bp-ICAM Annual Conference, Manchester, UK.
- **2023 - “Modelling CoO and Co₃O₄ nanoparticles for Fischer-Tropsch catalysts”** - bp-ICAM Annual Conference, Manchester, UK.
- **2023 - “Capturing the ground state electronic and magnetic properties of Cobalt containing metal and mixed metal oxides using density functional theory (DFT)”** - Materials Chemistry Consortium Annual Conference, Daresbury Laboratory, UK.

NATIONAL LEVEL EXAMINATIONS

- **Joint Admission Test for Masters (JAM) 2020 - Chemistry**
All India Rank: 216
Number of candidates appeared: 16,687
- **Graduate Aptitude Test in Engineering (GATE) 2021 – Chemistry**
All India Rank: 614
Number of candidates appeared: 23,465
- **Graduate Aptitude Test in Engineering (GATE) 2022 – Chemistry**
All India Rank: 248
Number of candidates appeared: 26,925

COURSES AND CERTIFICATIONS

- Advanced Learning Algorithms - DeepLearning.AI
- Mathematics for Machine Learning: Multivariate Calculus – Coursera
- Mathematics for Machine Learning: Linear Algebra – Coursera
- Supervised Machine Learning: Regression and Classification – Coursera
- Data Science: R Basics – EdX
- Android App Development – Internshala
- Python Programming: Basic to Advanced - Udemy